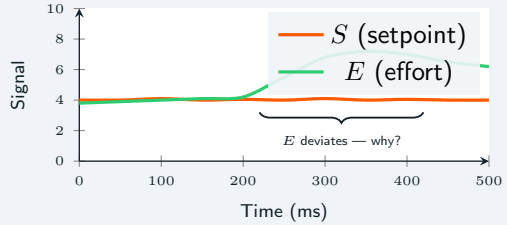
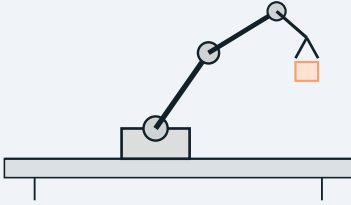


L1 · Seeing (Association) - What is?

$$P(E \mid S, C)$$

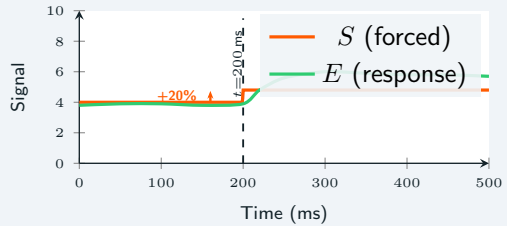
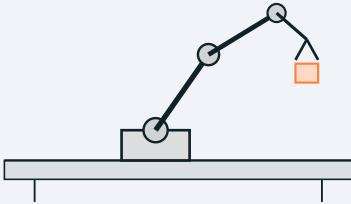


"I observe that when the payload is heavier, motor current is higher."

Correlation from passive observation — no action taken.

L2 · Doing (Intervention) - What if I do X?

$$P(E \mid \text{do}(S))$$

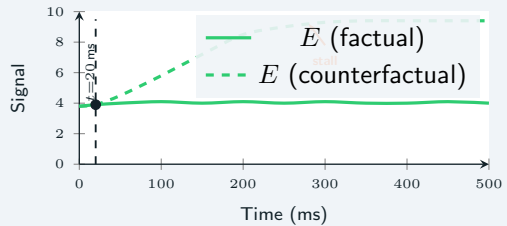
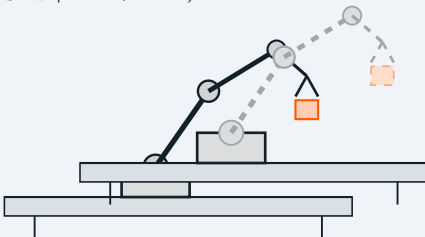


"If I command +20% torque now, the motor current will rise to ~X A."

Predicting the effect of an action at the present timestep.

L3 · Imaging (Counterfactual) - What if I had acted differently?

$$P(E_{S=x} \mid S=s, E=e)$$



"Given what actually happened, if torque had been raised to X at t = 200ms instead, the arm would have stalled here."

Reasoning over alternative histories, conditioned on what we observed.